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Premium

Catego
ry

Q1 How
How m

many

PAC

Assignment

address lines are required to
address

2MB of memory (byte-addressable) Memory size =

$$\underline{2\text{MBU}} = 2 \times 1024 \times 1024 \text{ bytes}$$

$$= 2,097,152 \text{ bytes}$$

since the memory is byte-addressable,
 each byte has
 a unique address

To find **how many** address lines are
 required,

Mamula We

use the formula?

22 = Number of addressable memory
 locations. So,

$$2^n = 2,097,152 = 2^{21} \Rightarrow \underline{\log_2(2,097,152)} = 21$$

→ 21 address lines are

Q2 2's complement
 data

required

VISION

A: 01010

MSB is 0

B: 11010

→ Positive

• Binary 91010 —

Decimal +10

B = 11010

- MSB is 1 Negative (2's complement)

To find value :

12 Invert & 00101

22 Add : 00101+1 = 00110 = 6

So B = -6

|(a) A+B = 10+(-6)=4

01010+11010 - 100100

1

1.1

010

11.010

$\pm$ 00 10 0 $\leftarrow$ Ignasing
carry out

$\rightarrow 00100.$

=

0 $\leftarrow$

4 in decimal

$\therefore$ No overflow
occurs

$$\underline{10} + 6 = 16$$

b) $A - B = (10) - (-6) = 10 + 6$

$$A - B =$$

$$B = 11010$$

$$A + (-B) .$$

$$A +$$

$$\text{Decimal} - 6$$

To find $-B$, we will
take

we will take o's
complement

- Invert the bits : 00101

Add 1

: 00101 + 00001 = 00110 (6 in
decimal)

$$A - B \equiv 01010 + 00110 = 10000$$

1

4 ±

+0011 10

Ominad

-10000 F 16 in decimal.

So, 17 is out of range of
representable values
for

5 bitsh

It's complement which is (~18) to
(15)

&

Overflow

occured.

$$COBA = B + (-A)$$

$$A = 01010 = \varepsilon + lo$$

$$-A = -10 =$$

add 1

B

B :

$$\text{Invert } 01010 = 10101$$

1010

-A

OLLO

110000

I

8. No

dD2*B

Ingering
carry bit

No

overflow

оссиha

B=-6 →→ Bino

Binary

211010

911010 y 1

Double it Using binary
addition

11010
+1101 © 2
10101 0

growing carry
bit.

Result : 10100 →2's

complement

||

• Invert 10100 = 01011; Add 1 → 01100-12
so value 13 -12

& No Overflow occurs

Vs

3 hist the assembly planguage progum
(of the

equivalent binary instruction)
generated by

compile

r femy

the

a

follow. J45, The
program

If statement: If $A-B$, 16,
30,

branch a statement 15% of $A-B$ to statement
BO of $A-B=0$; and to statement 45 if $A-B \neq 0$

100

ORG 100 || start of program at
memory co
10011

LOA

LDA

SUB B

STA

A

R

11 Load value

A into accumulat

11 Subtract $B-ACC=A-B$ #1 Store the result in R

11 **Reset** all flags
to 0

π load a

ViSiON

LDA Z

Jaschi

STA NEG

11 set Neg =

6

STA чено

11 Set Zero = 0

STA

POS

41 set POS = 0

1-zero check

LDA R

// Load result A-13

STA

|| 11 store in I

LOA I

11 Load I

ADD MI

117-1

ADD ONE || GT-1) +1 = T

SUB + || T-T=0 (only if I was Zero)

STA

T2 || store result in T2

|| Load result of T2

COA T2

STA ZERO || of T2=0 //

Negative Checks

+ Zero- remains 0 else I

CDA R IL Load result again ADD 7 11

Dummy Add

STA WER

11 store

neg

1/ Positive check LOA ONE

SUB

→ PUS=9-(NEG + ZERO)

| Load 1

NEG 11 subtract NEG

flag

SUB ZERO // subtract zero

flag

STA

ZeRo 11 Store of T2 =0 ; remains 0 eles: 1 Pos // To

NeG and zero are both 0

11 Halt

program

12

12 || first number

STA

MLT

A

·DEC

B

DEC 15

Z

DEC 0

MI

ONE DEC 1

DEC-1

R DEC

DEC

T

7.4 Processor

11 second number

4 constant zero

|| constant one

Constant minus

store result A-B Temp

for TORD Prunfor

24 R processor "w16 GRP's uses 32 bits

Instruction **format**, the instruction format consists of an op-code field an addressing mode field, register operand field, and a

16 Bits Scalar operand field if 8 addressing modes here to be

here to be supported the maximum number of unique op codes possible for every addressing mode is.

Total size = 32 bits

- Addressing Mode field

field · Needs 3 bits

(Because $2^3 = 8$ Addressing modes) .

- Register Operand field: Needs 4 bits

$2^4 = 16$ (GIRRS)
 $2^4 = 16$

VISION

- Special Operand field = Given as 18 bits

Total Used:

$3 \text{ (addressing mode)} + 4 \text{ (register operand)} + 16$
 $= 23 \text{ bits}$

Remaining
bits for

bits for op-code

$$\underline{32-23=19} \quad 32-23 = \underline{19}$$

bits = 29

$$2^9 \text{ bits} = 2^9 = 512$$

opcodes

∴ 512 unique opcodes possible
per addressing

mode

25: Consider a

→

3-stage, pipelined processor

having a delay of 15m (nanoseconds), 20ns,
17ns for the first second and third stages,

"defectively.

Assume that there is other

delay any predecessor does not
suffer

from the pipeline hazards. Also
assume that one Instruction is fetched
every cycle. The total execution time
for *executing* 100 instructions
is

Stage

1.

ns

delay : 15ns

Stage 2

delay Stage

3 delay

: 20ns

: 217ns

on this

proces
sor

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Number of Instruction =

100

- pipeline cycle
time:

stage

The slowest

determines

pipelines

clock cycle time

Pipeline

clocktime

$$= \max(\underline{15}, \underline{20}, 17) = 20 \text{ ns}$$

Total Execution Time

Total time

$$(K-1)n \times \text{clock}$$

$$(3-1) \times 100 \times 20$$

$$+ 100 \times 20$$

$$= 4000 \text{ ns}$$

& Total execution time

or 4000 nanoseconds they
will have

VISION

14-4